

Binary Search Halves the Problem

Searching a sorted list of 16 names for "Robert". Each check throws away half of what is left.

16 names still possible

Check 1 — middle is Henry. Henry comes before Robert, so drop the first eight.

8

Check 2 — middle is Leo. Drop four more.

4

Check 3 — middle is Nancy. Drop two more.

2

Check 4 — found, in four checks instead of sixteen.

Robert

1,000 names → linear search up to 1,000 checks • binary search 10
1,000,000 names → linear search up to 1,000,000 checks • binary search 20

This only works because the list is sorted. Sorting first is what buys the speed.